

Scattering by a Central Potential; the Hamiltonian — approved segments

19 approved segments, in reading order

scattering by a central potential

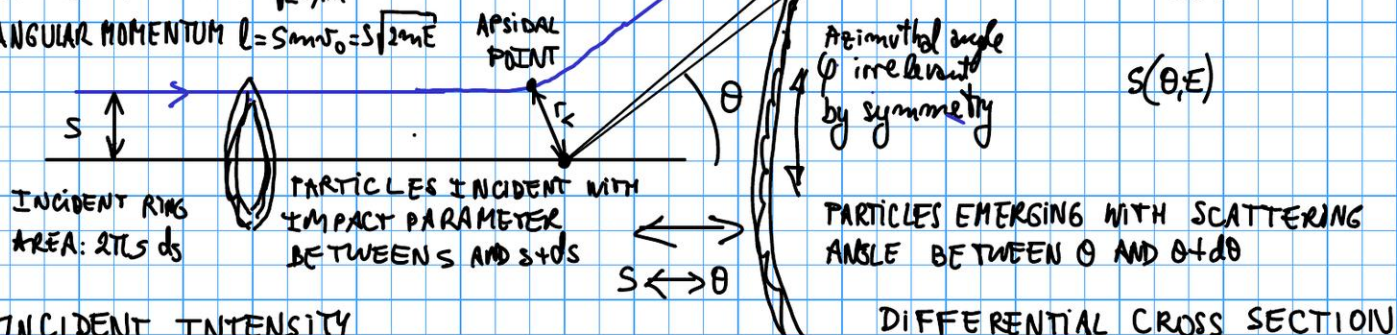
OCT 22 2024

INCIDENT ENERGY $E = \frac{m}{2} v_0^2$

INCIDENT VELOCITY $v_0 = \sqrt{2E/m}$

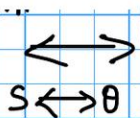
ANGULAR MOMENTUM $L = S \hbar \nu_0 = S \sqrt{2mE}$

$$S = \frac{l}{\sqrt{2mE}}$$



INCIDENT RING
AREA: $2\pi s ds$

IMPACT PARAMETER
BETWEEN s AND $s+ds$



PARTICLES EMERGING WITH SCATTERING
ANGLE BETWEEN θ AND $\theta+d\theta$

INCIDENT INTENSITY

$$I = \frac{\text{\# PART INCIDENT PER UNIT TIME}}{\text{INCIDENT BEAM CROSS SECTION AREA}}$$

DIFFERENTIAL CROSS SECTION

$$\sigma(\theta) d\theta = \frac{\text{\# PART SCATTERED INTO } d\theta \text{ PER UNIT TIME}}{\text{INCIDENT INTENSITY}}$$

$$I (2\pi s) |ds| = \text{\# part going through ring between } s \text{ and } s+ds = \text{\# part emerging with angle between } \theta \text{ and } \theta+d\theta = I \sigma(\theta) 2\pi \sin\theta |d\theta|$$

THEREFORE:

$$\sigma(\theta) = \frac{s}{\sin\theta} \left| \frac{ds}{d\theta} \right|$$

DEFINE ALSO THE TOTAL
CROSS SECTION:

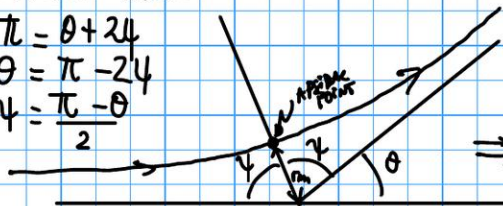
$$\sigma_T = \int d\Omega \sigma(\theta)$$

REPULSIVE CASE:

$$\pi = \theta + 2\psi$$

$$\theta = \pi - 2\psi$$

$$\psi = \frac{\pi - \theta}{2}$$



IN ALL
CASES:

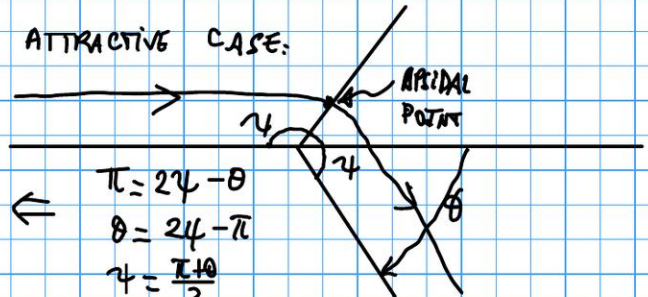
$$\theta = |\pi - 2\psi|$$

ATTRACTIVE CASE:

$$\pi = 2\psi - \theta$$

$$\theta = 2\psi - \pi$$

$$\psi = \frac{\pi + \theta}{2}$$



COMPUTING $S(\theta, E)$ (UNIVERSAL METHOD)

$$E = \frac{m}{2} \dot{r}^2 + V(r) + \frac{l^2}{2mr^2}$$

$$\dot{r} = \left[\frac{2}{m} \left(E - V(r) - \frac{l^2}{2mr^2} \right) \right]^{1/2}$$

$$\dot{r} = \frac{dr}{dt} \Rightarrow dt = \frac{dr}{\dot{r}}$$

$$l = mr^2 \dot{\theta} \Rightarrow d\theta = \dot{\theta} dt = \frac{l}{r^2} dt = \frac{l}{r^2} \frac{dr}{\dot{r}} = \frac{l}{\sqrt{m}} \frac{dr}{r^2 \sqrt{E - V(r) - \frac{l^2}{2mr^2}}}$$

$$E = \frac{m}{2} \dot{r}^2 + V(r) + \frac{l^2}{2mr^2} \quad \dot{r} = \left[\frac{2}{m} \left(E - V(r) - \frac{l^2}{2mr^2} \right) \right]^{1/2} \quad \dot{r} = \frac{dr}{dt} \Rightarrow dt = \frac{dr}{\dot{r}}$$

$$l = mr^2 \dot{\theta} \Rightarrow \frac{d\theta}{dt} = \dot{\theta} = \frac{l}{mr^2} \Rightarrow d\theta = \dot{\theta} dt = \frac{l}{mr^2} dt = \frac{l}{mr^2} \frac{dr}{\dot{r}} = \frac{l}{mr^2} \frac{dr}{\left[\frac{2}{m} \left(E - V(r) - \frac{l^2}{2mr^2} \right) \right]^{1/2}} = \frac{l}{m} \sqrt{\frac{m}{2E}} \frac{r^{-2} dr}{\left[1 - \frac{V(r)}{E} - \frac{l^2}{2mEr^2} \right]^{1/2}}$$

$$= \frac{l}{\sqrt{2EmE}} s = \frac{l}{\sqrt{2EmE}} s$$

$$\psi = \int_0^\psi d\psi = \int_{\theta_{\min}}^\pi d\theta = \int_{r_{\min}}^\infty \frac{s r^{-2} dr}{\left[1 - V(r) - s^2/r^2 \right]^{1/2}}$$

$$u = 1/r \quad du = -r^{-2} dr \quad u_{\min} = 1/r_{\min}$$

$$\Rightarrow \theta(s) = \left| \pi - 2 \int_0^{u_{\min}} \frac{s du}{\left[1 - \frac{V(1/u)}{E} - s^2 u^2 \right]^{1/2}} \right|$$

From here, compute $\left| \frac{ds}{d\theta} \right| = \frac{1}{\left| \frac{d\theta}{ds} \right|}$ and use it to get $\sigma(\theta) = \frac{s}{\sin \theta} \left| \frac{ds}{d\theta} \right|$

UP TO HERE, RESULTS ARE VALID FOR ANY CENTRAL POTENTIAL $V(r)$

Coulomb Scattering

$E > 0$	hyperbolic
$E = 0$	parabolic
$E < 0$	elliptic
ONLY FOR $V = -\frac{k}{r}$!!!	

FORCE $F = \frac{Z Z' e^2}{r^2}$

$f = -\frac{k}{r^2}$ $V = -\frac{k}{r}$
 $k = -Z Z' e^2$

Scattering: $E > 0$

$$\frac{1}{r} = C [1 + E \cos(\tilde{\theta} - \tilde{\theta}')] \quad C = \frac{mk}{l^2} = -\frac{m}{l^2} Z Z' e^2$$

$$\text{Eccentricity } E = \left[1 + \frac{2E l^2}{m k^2} \right]^{\frac{1}{2}} = \left[1 + \frac{2E l^2}{m (Z Z' e^2)^2} \right]^{\frac{1}{2}} = \left[1 + \left(\frac{2sE}{Z Z' e^2} \right)^2 \right]^{\frac{1}{2}} > 1$$

$$l = s m v_0 = s \sqrt{2mE} \rightarrow \frac{2l^2 E}{m} = \frac{s^2 2^2 m E^2}{m} = (2sE)^2$$

2 SITUATIONS: $\Rightarrow$ REPULSIVE CASE $Z Z' > 0 \Rightarrow k = -Z Z' e^2 < 0$

2 SITUATIONS: i) REPULSIVE CASE $zz' > 0 \Rightarrow k = -zz'e^2 < 0$
 ii) ATTRACTIVE CASE $zz' < 0 \Rightarrow k = -zz'e^2 > 0$ (ALSO GRAVITY $k = Gm_1m_2$)

(1) ATTRACTIVE CASE $ZE' < 0 \Rightarrow R = -ZE' > 0$ (ALSO GRAVITY $R = Gm_1m_2$)

i) REPULSIVE CASE: $k < 0$ $Q = \frac{mk}{b^2} < 0$

$$\frac{1}{r} = Q[1 + \epsilon \cos(\tilde{\theta} - \tilde{\theta}')] > 0 \Leftrightarrow 1 + \epsilon \cos(\tilde{\theta} - \tilde{\theta}') < 0 \Leftrightarrow \epsilon \cos(\tilde{\theta} - \tilde{\theta}') < -1 \Leftrightarrow \cos(\tilde{\theta} - \tilde{\theta}') < -\frac{1}{\epsilon}$$

We need the cosine to be negative and larger in magnitude than $1/\epsilon$

In this case, $\tilde{\theta} = \tilde{\theta}'$ is not a physical point, because $\cos(\tilde{\theta} - \tilde{\theta}') = \cos(0) = 1 \Rightarrow \frac{1}{r} = Q[1 + \epsilon] < 0$ UNPHYSICAL

SO $\tilde{\theta} = \tilde{\theta}'$ IS IN THE MIDDLE OF THE INACCESSIBLE REGION OF THE HYPERBOLA, AND $\tilde{\theta} = \tilde{\theta}' \pm \pi$ WILL BE THE PERIAPSIS, SO WE CHOOSE $\tilde{\theta}' = \pi$ AND WE GET

$$\cos(\tilde{\theta} - \tilde{\theta}') = \cos(\tilde{\theta} - \pi) = \cos \tilde{\theta} \underbrace{\cos \pi}_{-1} + \sin \tilde{\theta} \underbrace{\sin \pi}_0 = -\cos \tilde{\theta}$$

MAXIMUM AT $\tilde{\theta} = 0 \Rightarrow$

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$$\frac{1}{r} = \left(-\frac{mk}{l^2}\right) [-1 - \epsilon \cos(\tilde{\theta} - \tilde{\theta}')] = \frac{mZz'e^2}{l^2} [\epsilon \cos \tilde{\theta} - 1]$$

MAXIMUM AT $\tilde{\theta} = 0 \Rightarrow$

$$\text{MINIMUM } r: \frac{1}{r_m} = \frac{mZz'e^2}{l^2} [\epsilon - 1]$$

SO THE ANGLE AT THE APSIDAL POINT IS $\tilde{\theta} = 0$

$$\text{WE NOW NEED } \psi = \tilde{\theta}(r \rightarrow \infty) = \tilde{\theta}\left(\frac{1}{r} \rightarrow 0\right)$$

$$\left[\text{IN THE REPULSIVE CASE} \right]$$

$$\theta = \pi - 2\psi \Rightarrow \psi = \frac{\pi - \theta}{2}$$

$$0 = \epsilon \cos \psi - 1 \Leftrightarrow \frac{1}{\epsilon} = \cos \psi = \cos\left(\frac{\pi - \theta}{2}\right) = \underbrace{\cos \frac{\pi}{2}}_0 \cos \frac{\theta}{2} + \underbrace{\sin \frac{\pi}{2}}_1 \sin \frac{\theta}{2} = \sin \frac{\theta}{2}$$

$$\text{ii) ATTRACTIVE CASE: } k > 0 \quad C = \frac{k m}{l^2} > 0$$

$$0 = \epsilon \cos \psi - 1 \Leftrightarrow \frac{1}{\epsilon} = \cos \psi = \cos\left(\frac{\pi - \theta}{2}\right) = \underbrace{\cos \frac{\pi}{2}}_0 \cos \frac{\theta}{2} + \underbrace{\sin \frac{\pi}{2}}_1 \sin \frac{\theta}{2} = \sin \frac{\theta}{2}$$

ii) ATTRACTIVE CASE: $k > 0$ $C = \frac{k m}{\ell^2} > 0$

$$\text{TAKE } \tilde{\theta}' = 0 \quad \tilde{\theta} = \tilde{\theta}' = 0 \Rightarrow \frac{1}{r} = C[1 + \epsilon \cos(0)] = C[1 + \epsilon]$$

APSIDAL POINT IS ALSO AT $\tilde{\theta} = 0$

WE NOW NEED $\psi = \tilde{\theta}(r \rightarrow \infty) = \tilde{\theta}\left(\frac{1}{r} \rightarrow 0\right)$

MAX AT $\tilde{\theta} = 0$

MINIMUM r : $\frac{1}{r_m} = C[1 + \epsilon]$

[IN THE ATTRACTIVE CASE]
 $\theta = 2\pi - \pi \Rightarrow \psi = \frac{\pi + \theta}{2}$

$$0 = 1 + \epsilon \cos \psi \Leftrightarrow -\frac{1}{\epsilon} = \cos \psi = \cos\left(\frac{\pi + \theta}{2}\right) = \underbrace{\cos \frac{\pi}{2}}_0 \cos \frac{\theta}{2} - \underbrace{\sin \frac{\pi}{2}}_1 \sin \frac{\theta}{2} = -\sin \frac{\theta}{2}$$

We now combine both cases:

$$\cot^2 \frac{\theta}{2} = \frac{1}{\tan^2 \frac{\theta}{2}} = \frac{\cos^2 \frac{\theta}{2}}{\sin^2 \frac{\theta}{2}} = \frac{1 - \sin^2 \frac{\theta}{2}}{\sin^2 \frac{\theta}{2}} = \frac{1}{\sin^2 \frac{\theta}{2}} - 1 = \epsilon^2 - 1 = \left(\frac{2sE}{Z Z' e^2} \right)^2 \Rightarrow \boxed{s = s(\theta, \epsilon) = \left| \frac{Z Z' e^2}{2E} \right| \cot \frac{\theta}{2}}$$

$$\frac{ds}{d\theta} = \left| \frac{Z Z' e^2}{2E} \right| \frac{d}{d\theta} \left(\cot \frac{\theta}{2} \right) = \left| \frac{Z Z' e^2}{2E} \right| \left(-\frac{1}{2} \frac{\sin \frac{\theta}{2}}{\sin^2 \frac{\theta}{2}} - \frac{1}{2} \frac{\cos \frac{\theta}{2}}{\sin^2 \frac{\theta}{2}} \right) = - \left| \frac{Z Z' e^2}{4E} \right| \frac{1}{\sin^2 \frac{\theta}{2}} \quad \sigma(\theta) = \frac{s}{\sin \theta} \left| \frac{ds}{d\theta} \right| = \left| \frac{Z Z' e^2}{2E} \right|^2 \frac{1}{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}} \frac{1}{\sin^2 \frac{\theta}{2}}$$

$$\boxed{\sigma(\theta) = \frac{1}{4} \left(\frac{Z Z' e^2}{2E} \right)^2 \csc^4 \left(\frac{\theta}{2} \right)}$$

Rutherford scattering cross section
(for Coulomb forces)

Remarks:

- 1) The cross section is the same for the repulsive and attractive cases, since it is proportional to $(ZZ')^2$.

Remarks:

- 1) The cross section is the same for the repulsive and attractive cases, since it is proportional to $(zz')^2$.
- 2) It diverges like E^{-2} at low energies.
- 3) It diverges like θ^{-4} (VERY STRONG!!!) for small angles θ .
- 4) This classical result is exactly the same as the result obtained in Quantum Mechanics.

LITERATURE

4) This classical result is exactly the same as the result obtained in Quantum Mechanics.

Total scattering cross section:

$$\sigma_T = \int \sigma(\Omega) d\Omega$$

In this case $\sigma(\theta) = \text{const} \frac{1}{\sin^4(\frac{\theta}{2})}$

$$d\Omega = 2\pi \sin\theta d\theta = 4\pi \sin\frac{\theta}{2} \cos\frac{\theta}{2} d\theta$$

$$\sigma_T = \frac{4\pi}{4} \left(\frac{Zz'e^2}{2E} \right)^2 \int_0^\pi \frac{\sin\frac{\theta}{2} \cos\frac{\theta}{2}}{\sin^4\frac{\theta}{2}} d\theta$$

range: $0 < \theta \leq \pi$

NOTATION:

Goldstein

Others

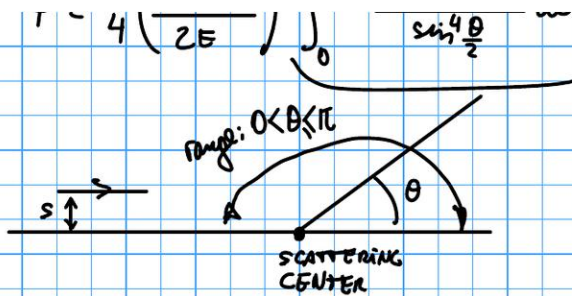
$$\sigma(\Omega) \leftrightarrow \frac{d\sigma}{d\Omega} \text{ DIFFERENTIAL SCATTERING CROSS SECTION}$$

$$\sigma_T \leftrightarrow \sigma \text{ TOTAL SCATTERING CROSS SECTION}$$

$$\sim \int_0^\pi \frac{d\theta}{(\theta/2)^3} = 8 \int_0^\pi \frac{d\theta}{\theta^3} = 8 \frac{\theta^{-2}}{(-2)} \Big|_0^\pi = +\infty \text{ STRONGLY DIVERGENT!}$$

CLASSICAL

$\sigma_T = +\infty$ for any long range potential



CLASSICAL

$\sigma_T = +\infty$ for any long range potential

QUANTUM

$\sigma_T = +\infty$ for some long range potentials
 $V(r)$ do not decay fast enough at $r \rightarrow \infty$

CAVEAT

1) IN A DENSE
 CONDUCTING MATERIAL:

$$V = -\frac{k}{r} \longrightarrow V(r) = -\frac{k}{r} e^{-r/\lambda} \quad \text{SCREENED COULOMB POTENTIAL ("YUKAWA")}$$

(In non-conducting materials $V^{\text{SCREEN}}(r)$ may take a different form, but still will vanish much faster than $1/r$)

Chapter 8 Hamiltonian and Hamilton's eqns of Motion

$$L = T - V = L(q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n, t)$$

$$0 = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} \quad p_i = \frac{\partial L}{\partial \dot{q}_i}$$

Legendre Transformation $q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n \rightarrow q_1, \dots, q_n, p_1, \dots, p_n$

MIDTERM REMARK

In polar coordinates

$$\frac{\partial L}{\partial \dot{\theta}} = p_\theta = l_z$$

$$L = L(r, \dot{r}, p_\theta)$$

Lagrange eqn

~~Lagrange Eqs~~

Legendre Transformation $q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n \rightarrow q_1, \dots, q_n, p_1, \dots, p_n$

$$f(x, y) \Rightarrow df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$$

In Thermodynamics

1st Law $dU = \delta Q + \delta W = T dS - p dV$ Fluid at constant N

$$U = U(S, V) \quad \text{Internal Energy}$$

We want T, V as variables

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$F = A = U - TS$ Helmholtz Free Energy

$$dA = dU - d(TS) = \cancel{T}dS - p dV - \cancel{T}dS - S dT = -S dT - p dV \Rightarrow A = A(T, V)$$

In general

$$f(x, y) = u dx + v dy \quad u = \frac{\partial f}{\partial x} \quad v = \frac{\partial f}{\partial y} \quad (x, y) \longrightarrow (u, y)$$

$$g = f - ux \Rightarrow dg = df - d(ux) = \cancel{u}dx + v dy - \cancel{u}dx - x du = -x du + v dy$$

$$g = g(u, y)$$

$$g = t - ux \Rightarrow dg = dt - d(ux) = \cancel{u}dx + vdy - \cancel{u}dx - xdu = -xdu + vdy$$

$$g = g(u, y)$$

let's look at $L(q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n)$

$$p_i = \frac{\partial L}{\partial \dot{q}_i} \bigg|_{\substack{q_j \\ \dot{q}_j}}$$

$$H = -\left(L - \sum_i p_i \dot{q}_i\right) \quad dH = -dL + \sum_i d(p_i \dot{q}_i) = -\sum_i \left(\frac{\partial L}{\partial q_i} dq_i + \frac{\partial L}{\partial \dot{q}_i} d\dot{q}_i \right) + \sum_i \left(\cancel{p_i} d\dot{q}_i + \dot{q}_i dp_i \right)$$

$$dH = -\frac{\partial L}{\partial t} dt + \sum_i \left(\dot{q}_i dp_i - \frac{\partial L}{\partial q_i} dq_i \right) \Rightarrow H = H(p_1, \dots, p_n, q_1, \dots, q_n, t)$$

$$dH = -\frac{\partial L}{\partial t} dt + \sum_i \left(\dot{q}_i dp_i - \frac{\partial L}{\partial q_i} dq_i \right) \Rightarrow H = H(p_1, \dots, p_n, q_1, \dots, q_n, t)$$

$$\boxed{\begin{aligned} \frac{\partial H}{\partial p_i} &= \dot{q}_i & \frac{\partial H}{\partial q_i} &= -\frac{\partial L}{\partial q_i} \stackrel{\text{Lagrange Eqn}}{=} -\dot{p}_i & \frac{\partial H}{\partial t} &= -\frac{\partial L}{\partial t} \end{aligned}}$$

CANONICAL EQS OF MOTION